

NTI — ADVANCED DATA ANALYSIS — BATCH 9

Healthcare Test Results Prediction

Final Project Report

A classification system predicting a patient's initial Test Result
(Normal / Abnormal / Inconclusive) at the moment of admission

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Phase 1 — Business Understanding

Business Objective

Build a classification system that predicts a patient's initial Test Result (Normal / Abnormal / Inconclusive) at admission time, using available patient and admission data, to help hospital staff prioritize medical follow-up.

Success Criteria

- Business KPI: improved prioritization of medical review for incoming patients.
- Technical Metric: classification accuracy above the random baseline (33.3% for 3 balanced classes).

Business Questions

1. Is there a relationship between Medical Condition and Test Results?
2. Does Admission Type (Urgent / Emergency / Elective) affect the test outcome?
3. Is Age a significant factor in test outcome?
4. Do Insurance Provider or Hospital show any distributional bias in outcomes?
5. Which classification model best predicts Test Results?

Dataset

health.csv — 55,500 records x 15 columns of patient admission data (demographics, medical condition, admission details, billing, and test results).

Team Roles

- Data Preparation
- Modeling
- Deployment (Streamlit)
- Power BI Dashboard
- Documentation

Phase 2 — Data Understanding

Dataset Structure

- 55,500 records x 15 columns.
- No missing values in any column.
- Column types: numeric (Age, Billing Amount, Room Number), categorical (Gender, Blood Type, Medical Condition, Admission Type, Insurance Provider, Test Results, Doctor, Hospital, Medication), dates (Date of Admission, Discharge Date), text (Name).

Data Quality Issues Identified

Issue	Finding
Duplicate rows	534 full-row duplicates
Negative Billing Amount	108 records with negative values (min ≈ -2008), invalid for a cost field
Name formatting	Inconsistent casing (e.g. "Bobby JacksOn")
Room Number	Numeric type but semantically categorical; not a useful predictive feature

Target Variable (Test Results)

Three classes, near-perfectly balanced: Abnormal 33.6%, Normal 33.4%, Inconclusive 33.1%. No class-imbalance issue for modeling.

Key Limitation — Documented Finding

Statistical checks (crosstabs of Medical Condition, Admission Type, Age vs. Test Results) show no meaningful relationship between any feature and the target — differences are within noise (under 1.5%). All categorical columns are near-uniformly distributed. This indicates the dataset is synthetically / randomly generated. Implication: modeling accuracy is expected to stay close to the random baseline (~33%), reported transparently as a data limitation rather than a modeling failure.

Phase 3 — Data Preparation

- Parsed Date of Admission and Discharge Date to proper datetime types.
- Dropped 534 duplicate rows.
- Cleaned text columns (Name, Doctor, Hospital): trimmed, standardized casing to Title Case.
- Engineered a new feature: Length of Stay (Discharge Date – Date of Admission).
- Removed rows where Discharge Date preceded Date of Admission (data integrity check).
- Fixed negative Billing Amount values by imputing the median billing amount per Medical Condition.
- Exported a clean, human-readable dataset for Power BI, and a separate encoded/scaled version (Label Encoding + StandardScaler) with an 80/20 stratified Train/Test split for Modeling.

Phase 4 — Exploratory Data Analysis

Business Question 1 — Medical Condition vs. Test Results

A crosstab (percentage by row) of Medical Condition against Test Results was visualized as a heatmap. Result: percentages across conditions are nearly identical (within ~1.5 points), showing no meaningful relationship between a patient's medical condition and their test outcome.

Business Question 2 — Admission Type vs. Test Results

Same crosstab approach applied to Admission Type (Urgent / Emergency / Elective). Result: outcome distribution is virtually the same across all admission types — admission type does not affect the test outcome.

Business Question 3 — Age vs. Test Results

Mean age was compared across the three outcome groups and visualized with a boxplot. Result: average ages differ by less than half a year across Normal, Abnormal, and Inconclusive groups — age is not a significant factor.

Business Question 4 — Insurance Provider / Hospital Bias

Crosstabs of Insurance Provider and of the top 10 Hospitals against Test Results were built. Result: no provider or hospital shows a meaningfully different outcome distribution — no evidence of bias.

EDA Conclusion

All four business questions point to the same conclusion: none of the available features shows a statistically meaningful relationship with Test Results. This confirms the Phase 2 finding that the dataset is synthetic, and sets the expectation that any classification model will perform close to the random baseline.

Phase 5 & 6 — Modeling and Evaluation

Features and Preprocessing

Features used: Age, Gender, Blood Type, Medical Condition, Insurance Provider, Admission Type, Medication. Target: Test Results. One-Hot Encoding applied to the 6 categorical features; Age passed through as-is. Data split 80/20 (train/test), stratified by target.

Models Compared

Model	Accuracy
Baseline (most-frequent class)	33.55%
Random Forest	38.49%
SVM (RBF kernel)	33.24%

Best Model

Random Forest performed best at 38.49% accuracy, about 5.2 points above the 33.3% random baseline. SVM essentially matched the baseline (no better than guessing).

Classification Report (Random Forest)

Precision, recall, and F1-score hover around 0.38–0.39 for every class — balanced across classes, but not strong. The confusion matrix shows errors spread fairly evenly across all three classes, consistent with the model picking up only weak, noise-level signal.

Sample Prediction

For a sample patient (Age 45, Female, Blood Type A+, Diabetes, Aetna insurance, Urgent admission, Aspirin), the Random Forest model produced a predicted Test Result with class probabilities, demonstrating the trained pipeline works end-to-end on new, unseen input.

Evaluation Conclusion

The Random Forest model modestly outperforms random guessing, which is the best achievable outcome given the documented lack of real relationships in the data. This result is consistent with, and confirms, the EDA findings in Phase 4 — the model and the exploratory analysis tell the same story.

Phase 7 — Deployment (Streamlit)

A Streamlit application was planned to expose the trained Random Forest pipeline as an interactive tool: input widgets for the 7 model features, on-demand prediction using the saved pipeline, a results/visualization tab reusing the Phase 4 charts, a requirements.txt listing dependencies, and deployment via Streamlit Community Cloud from a public GitHub repository, with any secrets kept out of the code.

Phase 8 — Power BI Dashboard

An additional deliverable requested by the instructor: a Power BI dashboard presenting the same findings visually, built on the cleaned dataset exported from Phase 3.

Data Model

- Fact table: cleaned admissions data (health data after cleaning).
- Dim_Date: a calendar table spanning Date of Admission to Discharge Date, generated with DAX CALENDAR(); active relationship on Date of Admission, inactive relationship on Discharge Date (activated per-measure via USERELATIONSHIP).
- Dim_Patient: deduplicated on Name, Age, Gender, and Blood Type, with a generated surrogate Patient ID. Documented limitation: the source data has no real Patient ID, so identical combinations of these four fields are treated as the same patient — an acceptable approximation given the synthetic dataset.
- Doctor and Hospital were kept as flat attributes in the fact table (not split into separate dimensions) to keep the model simple within the project timeline.

Report Pages

Page	Contents
Overview	KPI cards (Total Patients, Total Admissions, Total Billing, Avg Length of Stay, Avg Patient Age); Test Results donut; Admissions by Medical Condition; Admissions trend; Admissions by Admission Type

Page	Contents
Medical & Test Results Analysis	KPI cards (Total Doctors, Total Hospitals, Normal Results %, Avg Patient Age); Avg Length of Stay by Medical Condition; Gender split; Condition x Test Result matrix
Financial & Operations	KPI cards (Max/Min Billing Amount); Billing by Medical Condition; Billing by Hospital (Top 10); Billing by Insurance Provider; Billing by Month

Color Palette

Primary accent teal (#1D9E75) for icons, highlights and the cover slide; semantic colors for Test Results — green (#639922) for Normal, amber (#BA7517) for Inconclusive, red (#A32D2D) for Abnormal — applied consistently across all pages so the result category is recognizable at a glance.

Documented Dashboard Limitations

- Total Doctors (≈34K) and Total Hospitals (≈33K) are close to the total patient count (40K) because the synthetic data generator assigns a near-unique name per row; these counts do not reflect a realistic number of real-world providers and are reported as a data-quality observation rather than a business insight.
- Time-based charts (admissions trend, billing by month) aggregate across multiple years (2019–2024) unless filtered by the Year slicer, and should be read with that in mind.

Phase 9 — Documentation & Presentation

Summary of Findings

Medical Condition, Admission Type, Insurance Provider, and Hospital all show no meaningful relationship with Test Results at the population level (all within ~1–2 points of the 33.3% random baseline). Age is not a significant factor either. These findings, drawn independently from EDA, Power BI, and the trained model, are mutually consistent.

Recommendation

Given the confirmed synthetic nature of the dataset, any production use of this pipeline would require re-validation on real hospital records before deployment. The project nonetheless demonstrates a complete, correctly-executed analytical workflow: business framing, data quality auditing, cleaning, EDA, modeling, evaluation, dashboarding, and honest documentation of a negative result — which is itself a valid and important data-science outcome.

Deliverables

- Jupyter Notebook: full Python pipeline (cleaning → EDA → modeling → evaluation).
- Power BI file (.pbix): 3-page interactive dashboard.
- Streamlit application plan for live predictions.
- Final presentation deck (15 slides) and this project report.